Mega Project Synopsis on

DermaNova: AI-Driven Facial Skin Analysis & Personalized Skincare Recommendation System

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Contents

1 Introduction 3

2 Literature Review 4

3 Problem Statement 5

4 Objectives 5

5 Scope 6

6 Proposed Work 7

6.1 Methodology 7

6.2 Requirements 9

6.2.1 Software 9

6.2.2 Hardware 9

6.2.3 Python Libraries 10

6.2.4 Models Used 10

6.2.5 Dataset Requirements 11

6.2.6 Evalution metrics 11

6.3 Schedule 12

6.3.1 Summary Schedule 12

6.3.2 Detailed Schedule 13

7 Limitations of the Existing System 13

8 Conclusion 14

9 References 15

1. Introduction

Skin health is an important aspect of overall well-being, with conditions such as acne, pigmentation, inflammation, and other visible skin abnormalities affecting individuals across different age groups. Early identification of visible skin conditions can help users take appropriate preventive measures; however, access to timely dermatological consultation may be limited due to cost, availability, or geographical constraints. Recent advances in computer vision and deep learning have demonstrated the potential of artificial intelligence for automated analysis and classification of dermatological images [1], [4].

Conventional AI-based dermatological systems primarily focus on identifying or classifying specific skin lesions from clinical or dermoscopic images. Although these approaches have shown promising results, their performance can be influenced by factors such as dataset diversity, image quality, and variations in imaging conditions [2], [5]. There is therefore a need for user-oriented systems that can perform preliminary facial skin analysis using commonly available images while presenting the results in an understandable manner.

DermaNova is proposed as an AI-assisted facial skin analysis and personalized skincare guidance framework. The system allows users to upload or capture a facial image, performs image quality assessment and preprocessing, and applies a CNN-based deep learning model to classify supported skin-condition categories. Based on the predicted condition and relevant user information, the system generates personalized skincare guidance and stores previous analysis results for longitudinal skin-health tracking.

The proposed framework integrates computer vision, deep learning-based classification, personalized recommendation generation, and skin-health tracking into a unified platform. It is designed to provide preliminary AI-assisted information and educational skincare guidance while clearly communicating that its predictions and recommendations do not replace professional dermatological diagnosis, consultation, or treatment.

2. Literature Review

Deep learning has demonstrated considerable potential for automated dermatological image analysis. **Esteva *et al.*** showed that a deep convolutional neural network could achieve dermatologist-level performance for selected skin-cancer classification tasks, establishing the feasibility of AI-assisted dermatological image analysis [1].

Tschandl *et al.* introduced the **HAM10000 dataset** containing 10,015 dermatoscopic images across seven diagnostic categories, providing an important benchmark for machine-learning-based skin-lesion classification [2]. **Codella *et al.*** further contributed standardized ISIC challenge tasks for lesion segmentation, feature detection, and disease classification, enabling comparative evaluation of automated skin-image analysis methods [3].

Brinker *et al.* demonstrated the potential of CNN-based melanoma classification by comparing a deep-learning model with 157 dermatologists, reporting higher average sensitivity and specificity for the evaluated dermoscopic task [4]**. Jeong *et al.*** reviewed 65 deep-learning studies in dermatology and highlighted challenges related to variations in datasets, image types, model approaches, and evaluation metrics, which affect the translation of AI systems into practical clinical applications [5].

Kassem *et al.* also identified challenges including limited datasets, image-selection issues, and potential racial bias in automated skin-lesion classification systems [6]. These findings indicate the importance of representative datasets, robust evaluation, and real-world image diversity when developing AI-based dermatological systems.

Research Gap

Existing research predominantly focuses on skin-lesion classification, detection, and segmentation. There is scope for an integrated user-oriented system that combines facial skin analysis, personalized skincare guidance, and longitudinal skin-health tracking. **DermaNova** addresses this gap by integrating these components into an AI-assisted framework while maintaining that the system is intended for preliminary assistance and not as a replacement for professional dermatological diagnosis.

3. Problem Statement

Skin-related conditions such as acne, pigmentation, melasma, eczema, and other dermatological abnormalities affect millions of people worldwide. Early identification and proper skincare management play an important role in maintaining healthy skin; however, many individuals lack access to timely dermatological consultation or rely on generic skincare advice that may not be suitable for their specific skin characteristics. Existing AI-based skin analysis systems primarily focus on disease classification without integrating personalized skincare recommendations, user profile information, or long-term skin-health monitoring. Furthermore, variations in image quality, lighting conditions, skin tones, and dataset diversity reduce the reliability of automated skin analysis systems in real-world environments.

This project proposes **DermaNova, an AI-driven facial skin analysis** framework that performs image preprocessing, deep learning-based skin condition classification, personalized skincare recommendation, and skin-health tracking through a unified intelligent platform. The proposed framework aims to provide accurate preliminary skin analysis while assisting users with personalized skincare guidance and encouraging professional dermatological consultation whenever necessary.

4. Objectives

The main objectives of this project are:

1. **To develop an AI-powered facial skin analysis system** capable of identifying supported skin conditions from facial images using deep learning techniques.

2. **To improve prediction reliability by applying image preprocessing techniques** such as image enhancement, resizing, normalization, and noise reduction before classification.

3. **To implement a deep learning-based skin condition classification model** using Convolutional Neural Networks (CNN) and transfer learning techniques for accurate prediction.

4. **To generate personalized skincare recommendations** by combining predicted skin conditions with user profile information, skin characteristics, and skincare preferences.

5. **To develop a skin-health tracking module** that stores previous analyses, enabling users to monitor changes in their skin condition over time.

6. **To evaluate the proposed framework using standard machine learning performance metrics** such as Accuracy, Precision, Recall, F1-Score, ROC-AUC (where applicable), and Confusion Matrix while maintaining appropriate medical safety boundaries.

5. Scope

This research focuses on developing an intelligent AI-assisted facial skin analysis and personalized skincare recommendation framework for preliminary assessment of common facial skin conditions. The proposed system utilizes image preprocessing and deep learning techniques to analyze facial skin images captured using standard digital cameras or smartphones. The framework performs automated feature extraction and classification to identify supported skin-condition categories and subsequently generates personalized skincare guidance based on the predicted condition and userprovided profile information.

In addition to skin analysis, the system maintains historical analysis records to facilitate skin-health tracking and progress monitoring over time. The proposed framework is intended for educational and decision-support purposes and is **not designed to replace professional dermatological diagnosis or treatment.** The system can be further extended to support additional skin conditions, multilingual interfaces, mobile deployment, dermatologist integration, and larger datasets representing diverse skin tones, particularly those relevant to the Indian population.

6. Proposed work

The proposed work aims to develop an integrated AI-powered platform that combines computer vision, deep learning, and personalized recommendation techniques to assist users in understanding their facial skin condition. The system accepts facial images uploaded or captured through a web application, preprocesses the images to improve quality, and performs automated skin condition classification using a trained CNN-based deep learning model.

Following classification, the predicted skin condition is combined with user profile information such as skin type, age group, skincare preferences, and previous analysis history to generate personalized skincare guidance. The framework also stores historical records, allowing users to monitor changes in skin health over time through comparative analysis. By integrating image preprocessing, AI-based classification, recommendation generation, and progress tracking into a single platform, DermaNova aims to provide an intelligent and accessible solution for AI-assisted skincare management while emphasizing that professional medical consultation remains essential for diagnosis and treatment.

6.1 Methodology

The proposed methodology of **DermaNova** begins with acquiring a facial skin image through a web-based interface using an uploaded image or smartphone/webcam capture. The system performs image quality assessment based on factors such as resolution, brightness, focus, and face visibility. The accepted image is then preprocessed using face detection, cropping, resizing, illumination normalization, noise reduction, contrast enhancement, and pixel normalization.

The preprocessed image is analyzed using a **CNN-based deep learning model**, such as EfficientNet or ResNet, trained on dermatological image datasets. The model classifies the image into supported skin-condition categories and generates a confidence score. Low-confidence predictions are flagged for professional dermatological consultation. The prediction is then combined with user information such as skin type, age group, allergies, preferences, and previous analysis to generate personalized skincare guidance.

The system stores the analysis results, confidence scores, recommendations, and timestamps for **skin-health tracking**. Model performance is evaluated using **Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix**. Thus, DermaNova integrates image processing, deep learning-based classification, personalized recommendations, and skin-health tracking into a unified AI-assisted facial skin analysis framework.

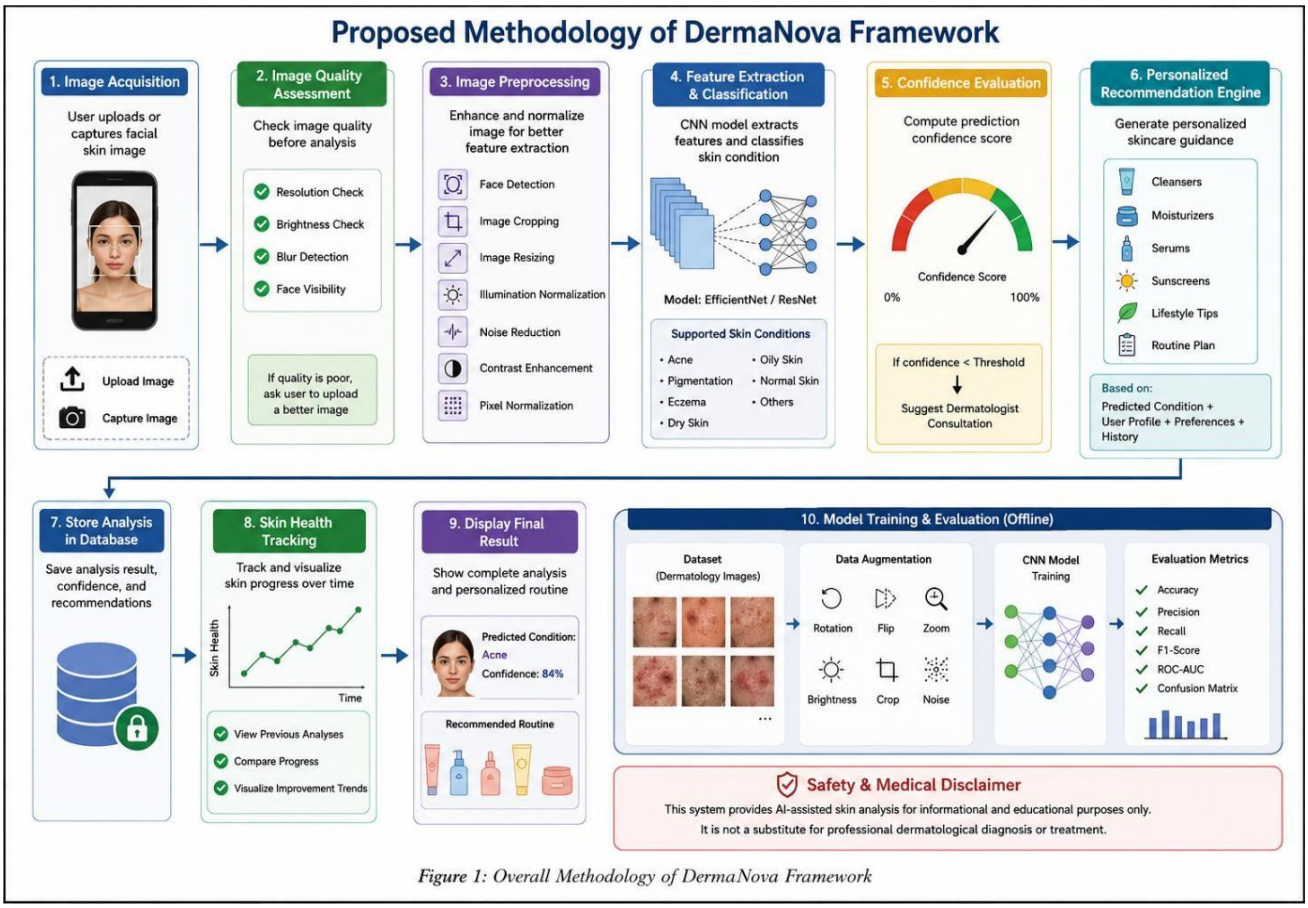


Figure 1: Overall Methodology of DermaNova Framework

Figure 1 : Overall Methodology of DermaNova Framework

6.2 Requirements

6.2.1 Software Requirements

The proposed DermaNova framework requires the following software components:

- **Operating System:** Windows 10/11 or Ubuntu Linux
- **Programming Language**: Python 3.11+
- F**rontend Framework:** Next.js
- **Backend Framework:** FastAPI
- **Database:** MongoDB
- **IDE:** Visual Studio Code
- **Version Control:** Git & GitHub
- **Model Development**: Google Colab / Jupyter Notebook
- **API Testing:** Postman
- **Deployment Platform:** Vercel / Railway

6.2.2 Hardware Requirements

The recommended hardware configuration is as follows:

- **Processor:** Intel Core i5 (10th Generation) or higher
- **RAM:** Minimum 8 GB (16 GB Recommended)
- **Storage:** 256 GB SSD or higher
- **GPU:** NVIDIA RTX 3050 or equivalent (recommended for model training)
- **Camera**: Smartphone Camera or HD Webcam
- **Internet Connection**: Broadband Connection

6.2.3 Python Libraries

The implementation utilizes the following Python libraries:

- **TensorFlow / Keras** – Deep Learning Model Development
- **OpenCV** – Image Processing
- **NumPy** – Numerical Computation
- **Pandas** – Dataset Management
- **Scikit-learn** – Model Evaluation
- **Matplotlib** – Data Visualization
- **Pillow (PIL)** – Image Handling
- **FastAPI –** Backend API Development
- **Uvicorn** – ASGI Server
- **PyMongo** – MongoDB Connectivity

6.2.4 Models Used

The proposed system utilizes deep learning architectures for skin-condition classification.

- **Convolutional Neural Network (CNN):** Primary architecture for image classification.
- **EfficientNet-B0**: Selected as the primary model due to its high accuracy and computational efficiency.
- **ResNet-50:** Used for performance comparison.
- **MobileNetV3:** Suitable for lightweight deployment.
- **DenseNet121:** Evaluated as an alternative deep learning architecture.

6.2.5 Dataset Requirements

The proposed framework requires high-quality dermatological image datasets.

- **HAM10000** Dataset
- **ISIC Archive**
- **DermNet** (subject to licensing)
- Future Indian Skin Dataset (if available)

The dataset will undergo:

- Image Quality Assessment
- Image Resizing
- Image Normalization
- Data Augmentation
- Label Verification
- Training, Validation, and Testing Split

6.2.6 Evaluation Metrics

The performance of the proposed framework will be evaluated using:

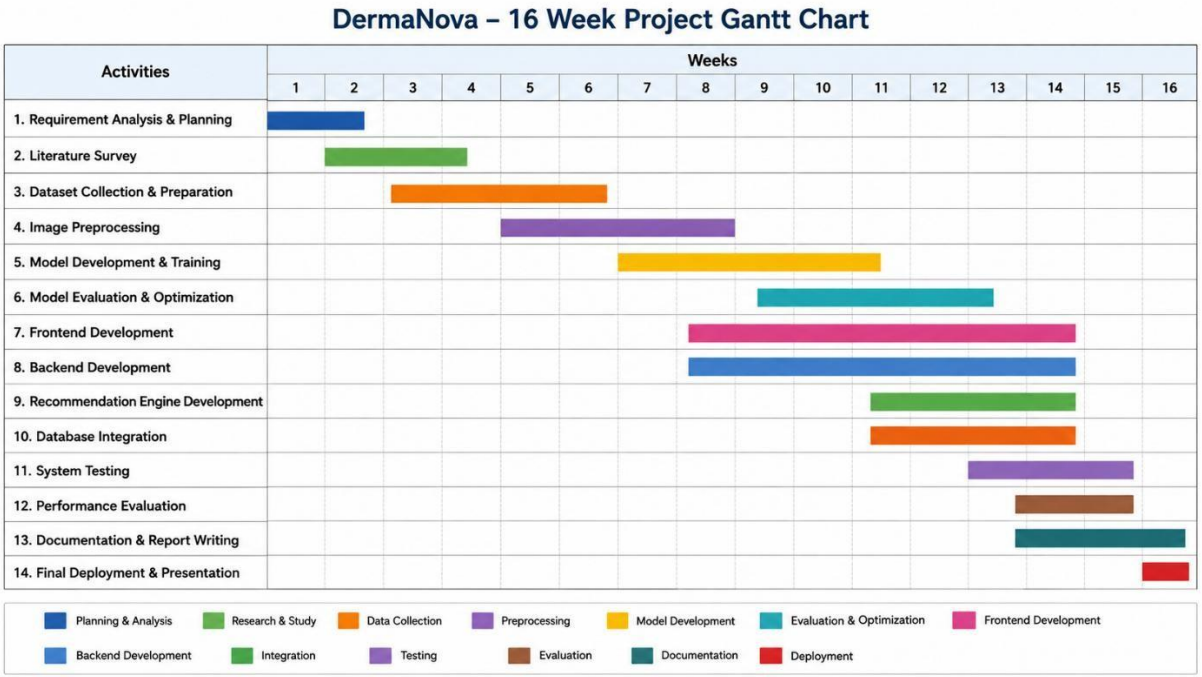
- **Accuracy**
- **Precision**
- **Recall (Sensitivity)**
- **F1-Score**
- **ROC-AUC**
- **Confusion Matrix**
- **Inference Time**

6.3 Schedule

The development of the proposed DermaNova framework is planned over 16 weeks. The project is divided into multiple phases, including requirement analysis, literature survey, dataset collection, model development, frontend and backend implementation, system integration, testing, documentation, and final deployment. Each phase is designed to ensure systematic progress while allowing sufficient time for experimentation, evaluation, and refinement of the AI model. The proposed schedule ensures balanced distribution of work among team members and provides adequate time for model optimization, application development, testing, and report preparation.

6.3.1 Summary Schedule

- **Phase I**: Requirement Analysis and Literature Survey
- **Phase II:** Dataset Collection and Image Preprocessing
- **Phase III:** Deep Learning Model Development and Training
- **Phase IV:** Frontend and Backend Development
- **Phase V:** System Integration and Recommendation Engine Development
- **Phase VI**: Testing, Evaluation, and Performance Analysis
- **Phase VII:** Documentation, Research Paper Preparation, and Final Deployment



6.3.2 Detailed Schedule

- **Week 1–2:** Requirement Analysis and Project Planning
- **Week 2–4:** Literature Survey and Technology Study
- **Week 3–6:** Dataset Collection and Preparation
- **Week 5–8:** Image Preprocessing and Data Augmentation
- **Week 7–11:** Model Development and Training
- **Week 9–12:** Model Evaluation and Optimization
- **Week 8–14:** Frontend and Backend Development
- **Week 11–14:** Recommendation Engine and Database Integration
- **Week 13–15:** System Testing and Performance Evaluation
- **Week 14–16:** Documentation and Report Preparation
- **Week 16:** Final Deployment and Project Presentation

7. Limitations of Existing System

Existing skin-analysis and skincare systems have several limitations:

- Many systems provide **generic recommendations** without considering individual skin characteristics.
- Most AI systems focus mainly on **skin-condition classification** rather than personalized skincare guidance.
- Prediction accuracy can be affected by **image quality, lighting, camera resolution, and viewing angle**.

- Available dermatological datasets may have **limited skin-tone diversity**, affecting generalization across populations.
- Most systems lack **continuous skin-health tracking** and comparison of previous results.
- Some solutions require **dermoscopic images**, limiting usability with ordinary smartphone images.
- Existing platforms often lack integration of **AI analysis, personalization, user profiles, and recommendations** in a single system.
- AI predictions may be inaccurate for **complex or uncommon conditions** and therefore cannot replace professional dermatological diagnosis

8. Conclusion

The proposed **DermaNova** framework integrates computer vision, deep learning, and recommendation techniques to provide preliminary facial skin analysis and personalized skincare guidance. The system analyzes facial images using preprocessing and CNN-based classification, combines the results with relevant user information, and maintains previous analysis records for skin-health tracking.

The performance of the proposed model will be evaluated using **Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix**. By integrating analysis, personalization, and tracking within a single platform, DermaNova aims to provide users with accessible and informative AI-assisted skincare support.

The system is intended as a **decision-support and educational tool** and does not replace professional dermatological diagnosis or treatment. Future enhancements can include **larger and more diverse datasets, improved skin-tone representation, explainable AI, mobile deployment, and integration with professional healthcare services**.

9. References

1. **[1]** A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, pp. 115–118, 2017. **Available at:** https://doi.org/10.1038/nature21056

2. **[2]** P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," *Scientific Data*, vol. 5, Art. no. 180161, 2018. **Available at:** https://doi.org/10.1038/sdata.2018.161

3. **[3]** N. C. F. Codella, D. Gutman, M. E. Celebi, B. Helba, M. A. Marchetti, S. W. Dusza, A. Kalloo, K. Liopyris, N. Mishra, H. Kittler, and A. Halpern, "Skin Lesion Analysis Toward Melanoma Detection: A Challenge at the 2017 International Symposium on Biomedical Imaging (ISBI), Hosted by the International Skin Imaging Collaboration (ISIC)," *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2017. **Available at:** https://arxiv.org/abs/1710.05006

4. **[4]** T. J. Brinker, A. Hekler, A. H. Enk, J. Klode, A. Hauschild, C. Berking, B. Schilling, M. Haferkamp, and others, "Deep learning outperformed 136 of 157 dermatologists in a head-to-head dermoscopic melanoma image classification task," *European Journal of Cancer*, vol. 113, pp. 47–54, 2019. **Available at:** https://doi.org/10.1016/j.ejca.2019.04.001

5. **[5]** H. K. Jeong, C. Park, R. Henao, and M. Kheterpal, "Deep Learning in Dermatology: A Systematic Review of Current Approaches, Outcomes, and Limitations," *JID Innovations*, vol. 3, no. 1, Art. no. 100150, 2023. **Available at:** https://doi.org/10.1016/j.xjidi.2022.100150

6. **[6]** M. A. Kassem, K. M. Hosny, R. Damaševičius, and S. E. M. El-Ghamry, "Machine Learning and Deep Learning Methods for Skin Lesion Classification and Diagnosis: A Systematic Review," *Diagnostics*, vol. 11, no. 8, Art. no. 1390, 2021. **Available at:** https://doi.org/10.3390/diagnostics11081390

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